ATHARVA PATIL

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PROFESSIONAL SUMMARY

Final-year AI/ML Engineering student with strong Python and SQL skills, and hands-on experience across machine learning, time-series forecasting, and data-driven decision-making — including an agricultural commodity price forecasting system delivering 28-day forecasts for five commodities. Skilled in feature engineering, model optimization, and deploying models via FastAPI, with proven ability to present technical findings through interactive dashboards. Seeking a Data Science internship to apply strong analytical fundamentals to real-world manufacturing and supply chain challenges.

EDUCATION

M.M. College of Engineering, Pune, Maharashtra Expected Graduation: Jul 2027

Bachelor of Engineering (B.E.) in Artificial Intelligence & Data Science — CGPA: 8.5/10

EXPERIENCE

AI/ML Developer Intern — CAD Automation Jan 2026 – Mar 2026

MCAD Solution

- Developed a rule-based engineering drawing validation system in Python using **OCR (Tesseract)** to detect dimension mismatches and validate manufacturing standards from PDF drawings, reducing manual review effort.
- Automated generation of 2D engineering drawings from 3D CAD models using the **SolidWorks API**, **COM Automation (win32com.client)**, and **VBScript** macros, streamlining CAD-to-drawing workflows.
- Implemented automated orthographic view generation (Front, Top, Right, Isometric) and built **Tkinter**-based GUI tools, improving drawing standardization and accessibility across **CATIA V5** and **SolidWorks** environments.

PROJECTS

Agri-Commodity Price Forecasting System *Jul 2025*

- Engineered a complete **time-series machine learning** pipeline with advanced **feature engineering** and **model optimization** to forecast agricultural commodity prices.
- Delivered **28-day price forecasts** for **five commodities** (Tomato, Onion, Cabbage, Brinjal, Potato), enabling data-driven market insights and improved price-stability planning.
- **Achievement:** Participated in InnoHack 2.0, VIT (Jul 2025).

Krishi Mitra AI — Plant Disease Detection System *Oct 2025*

- Engineered a **MobileNetV2**-based **computer vision** model achieving **~90% accuracy** across **12,000+ images**, applying **data augmentation** and **class-weight balancing** to resolve dataset imbalance.
- Built and deployed a scalable **FastAPI** inference pipeline converting model probabilities into actionable diagnostics, assigning dynamic risk levels (Low/Medium/Critical) across **9 disease categories** for real-time decision support.
- **Achievement:** Secured 2nd Place at IEEE SB MMCOE – INNOVERSE (Oct 2025).

Intelligent Customer Segmentation Dashboard *Sep 2025*

- Built an **unsupervised machine learning** pipeline using **RFM (Recency, Frequency, Monetary)** analysis to segment e-commerce customers, benchmarking five clustering algorithms (**K-Means**, **GMM**, **DBSCAN**, **Agglomerative**, **BIRCH**) and identifying **GMM** as top performer with a **Silhouette Score of 0.80**.
- Designed an interactive **Streamlit** dashboard leveraging **PCA** for high-dimensional visualization, surfacing customer personas ("Champions", "At-Risk") for targeted marketing strategies.
- **Achievement:** Secured 3rd Place, Mindstorm 2K25, MMCOE (Sep 2025).

TECHNICAL SKILLS

Programming & Version Control: Python (Primary), SQL, MySQL, Git/GitHub, C++, VBScript

Machine Learning & AI: Supervised & Unsupervised Learning, K-Means, GMM, DBSCAN, Agglomerative Clustering, Random Forest, XGBoost, Time-Series Forecasting, Predictive Modeling, Computer Vision (CNN, MobileNetV2), Feature Engineering, Model Optimization

Model Deployment & Automation: FastAPI, RESTful APIs, API Integration, Automation Scripting

Data Analytics & Visualization: Exploratory Data Analysis (EDA), Power BI, Matplotlib, Seaborn, Excel, Streamlit, Google Sheets Automation

Core CS Fundamentals: Object-Oriented Programming (OOP), DBMS, Data Structures & Algorithms (DSA), Problem Solving

Tools & Platforms: SolidWorks API, CATIA V5, COM Automation (win32com.client), OCR (Tesseract), Tkinter, PDF Data Extraction, Macro & Workflow Automation

ACHIEVEMENTS

- **2nd Place, IEEE SB MMCOE – INNOVERSE (Oct 2025)**
- **3rd Place, Mindstorm 2K25, MMCOE (Sep 2025)**
- **Participant, InnoHack 2.0, VIT (Jul 2025)**